

Møller Polarimeter Slow Control

18 August 2004

File: slow_mpc.html

The Moller Polarimeter Control program (MPC) have been crated to control Moller Detector Settting, Colimator and Slide Position, High Voltage Setting and monitoring of the Epics Variables and Moller Target Status over network using TCP/IP connection. It has GUI (Main Window) to show applications for control or monitoring and runs server on port 5773 that accepts external requests. Every application can be started as stand-alone program. There are few scripts located in ~/Java/msetting/ to start applications:

- `einfo` — to start Epics Monitor program;
- `mhv` — to start High Voltage Control program;
- `ms` — to start Moller Detector Setting program;
- `motor` — to start Motor Control program;

The configuration parameters to start application are defined inside scripts. There is no server when applications running alone.

Start MPC program

To start MPC programm there is script with name “`mpc`”. After login with moller account change directory to `~moller/Java/msetting` and type “`mpc`”.

Properties file

The MPC program has configured by using properties file that located in the user directory with name “`MPC.props`”. This file stored names of application configuration files. Properties file is loaded to MPC program when it is started. The file has next format (example):

MPC.props example (Thu Feb 06 18:01:14 EST 2003)

Property	Value
<code>config.file4</code>	<code>config5</code>
<code>config.file3</code>	<code>/home/moller/Java/msetting/motorControl.config</code>
<code>config.file2</code>	<code>/home/moller/Java/msetting/hvControl.config</code>
<code>config.file1</code>	<code>/home/moller/mpset</code>
<code>config.file0</code>	<code>/home/moller/Java/msetting/epicsMonitor.config</code>
<code>server.port</code>	<code>5773</code>
<code>app.names</code>	<code>Epics Monitor:Detector Setting:High Voltage Control:Motor Control:Target Monitor:</code>

Where,

- `app.names` — names(title) of applications separated by “`:`” symbol;
- `server.port` — port number of the MPC server;
- `config.fileN` — full name(with path) of the configuration file for corresponded application;

Main Window

Main Window shows Menu Bar, and Status Window.

Menu bar includes next menu items: **File**, **View**, **Help**

File

includes following items:

- **Quit** — to exit from program with confirmation

View

includes follows menu items:

- **Epics Monitor** — to open new window with table, that shows information gathering from Epics using cdevGateway server. To start cdevGateway server login on ‘adaqs2’ and type “cdevGatewayStart”. The cdevGateway server port for connection is set now to 9573. The parameters to start of this application stored in configuration file that has next format:

Parameter	Value
gatewayhost	adaql1
gatewayport	9573
epicslist	/adaqfs/home/moller/daq/coda2/RunInfo/epics.list

The full name(with path) of configuration file stored in poperties file that located in user directory with name “MPC.props”. The configuration file name placed after property “config.file0”.

- **Detector Setting** — to open new window with panels to control and setting parameters of the Moller Detector Electronic Modules that installed in CAMAC crate over network using connection to the server. Server runs on the VME CPU. The VME server port for connection is set now to 5222. The parameters to start of this application stored in configuration file that has next format (example):

Date 04:57:29 PM Feb 11, 2003. hostname:port hal lavme5:5222

LeCroy-2365 PLU Module, slot 11, CTR 0

channel	andBits	orBits	Not	Logic
0	4105	0	1	not(not(0)+not(3)+not(12))
1	4114	0	1	not(not(1)+not(4)+not(12))
2	4123	0	1	not(not(0)+not(1)+not(3)+not(4)+not(12))
3	6226	0	1	not(not(1)+not(4)+not(6)+not(11)+not(12))
4	4128	0	1	not(not(5)+not(12))
5	5120	0	1	not(not(10)+not(12))
6	6145	0	1	not(not(0)+not(11)+not(12))
7	4097	8192	1	not(not(0)+not(12)+(13))

TD8000 LedDiscriminator, slot 2, WIDTH 32

	1	2	3	4	5	6	7	8
MASK	0	0	0	0	0	0	0	0
THRESHOLD	39	39	80	80	80	900	900	900

LeCroy-4518 Delay Line, slot 6

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
DELAY	6	6	2	18	18	2	2	2	26	2	2	2	26	26	26	26

The full name(with path) of configuration file stored in poperties file that located in user directory with name “MPC.props”. The configuration file name placed after property “config.file1”.

- **High Voltage Contol** — to open new window with panels to control and setting parameters of the High Voltage Mainframe type of LeCroy-1450 over network using connection to the portserver and telnet protocol. The

parameters to start of this application stored in configuration file that has next format:

Parameter	Value
hvHost	hatsv5
hvPort	2011
login	user
passwd	xxxxx
slots	S5 S6
dataDir	/home/moller/daq/coda2/

The full name(with path) of configuration file stored in poperties file that located in user directory with name “MPC.props”. The configuration file name placed after property “config.file2”.

- **Motor Control** — to open new window with panels to remotely control of 2 elements: collimator and slide.
 - *Collimator* at the dipole entrance consists of 2 lead (or some other heavy material) bricks in each arm. The distance between the bricks is adjustable in a range of about 2-40mm. The collimators in both arms are moved at the same time, leaving a horizontal slit.
 - *Slide* is a movable platform positioned between the dipole and the detector box. It is used to test detectors. The detectors can be attached to this platform.

Both the collimator and the slide can be moved remotely using stepping motors controller “VELMEX-NF90”. The parameters to start of this application stored in configuration file that has next format (it is one line for one motor to control):

Fields, in order: Title, HostName, Port, Motor#, Acceleration, Speed, MinRange(steps), MaxRange(steps), PosCoefficient(steps/mm), LastPosition(steps).

Where,

- Title — is the name and title of control panel;
- HostName — is the name of the telnet portserver;
- Port — is port number of the telnet portserver for connection;
- Motor# — is the name of the motor to control (example: 1M);
- Acceleration — is value of acceleration parameter (type: long; range: 1 to 50);
- Speed — is value of speed parameter (type: long; range: 1 to 6000);
- MinRange — is minimum absolute value of position in steps (type: long);
- MaxRange — is maximum absolute value of position in steps (type: long);
- PosCoefficient — is coefficient(steps/mm) to recalculate position from steps to mm (type: float);
- LastPosition — is last stored absolute value of position in steps (type: long);

Example for two elements: Slide and Collimator:

Title	HostName	Port	Motor#	Acceleration	Speed	MinRange	MaxRange	PosCoefficient	LastPosition
Slide	hatsv12	2003	1M	5	1000	0	128350	633.3	0
Collimator	hatsv12	2002	1M	2	20	0	10000	250.0	2500

The full name(with path) of configuration file stored in poperties file that located in user directory with name “MPC.props”. The configuration file name placed after property “config.file3”.

Help

shows help file.

MPC Server

External requests

To communicate with MPC Server the program ‘mpcclient’ is used. This program gets as parameters the hostname (where MPC server is running), portnumber and message for sending to the MPC Server:

```
mpcclient hostname portnumber 'message to send'
```

For example, next command send command message “GET EPICSINFO” (get information from Epics) to the MPC Server that runs on *adaql1* host:

```
mpcclient adaql1 5773 'get epicsinfo'
```

The command message is string with ASCII characters. The format of this command message is follows:

<command> <property>

<command> — command word to server, one of follows:

- *GET* — get values of <property>. This command valid only for next values of <properties>:
 - EPICSINFO — get epics information;
 - DETECTORINFO — get current detector settings;